STAGE 4: DOCUMENTATION, INSIGHTS & PRESENTATION

SALES PERFORMANCE ANALYSIS AND PROFIT OPTIMIZATION OF NIKE USING DATA ANALYTICS

INTRODUCTION

This project focuses on analysing Nike's retail and online sales transactions to understand business performance, customer purchasing behaviour, and profitability trends. The dataset contains real-world–like sales records with several inconsistencies such as missing values, duplicate entries, incorrect formats, and logical errors. Therefore, the analysis begins with extensive data cleaning and preprocessing before performing exploratory data analysis (EDA) and visualization.

The main goal of this project is to transform raw and messy sales data into meaningful insights that help improve revenue, optimize discounts, identify high-performing products and regions, and support better business decisions through interactive dashboards.

DATA EXPLORATION

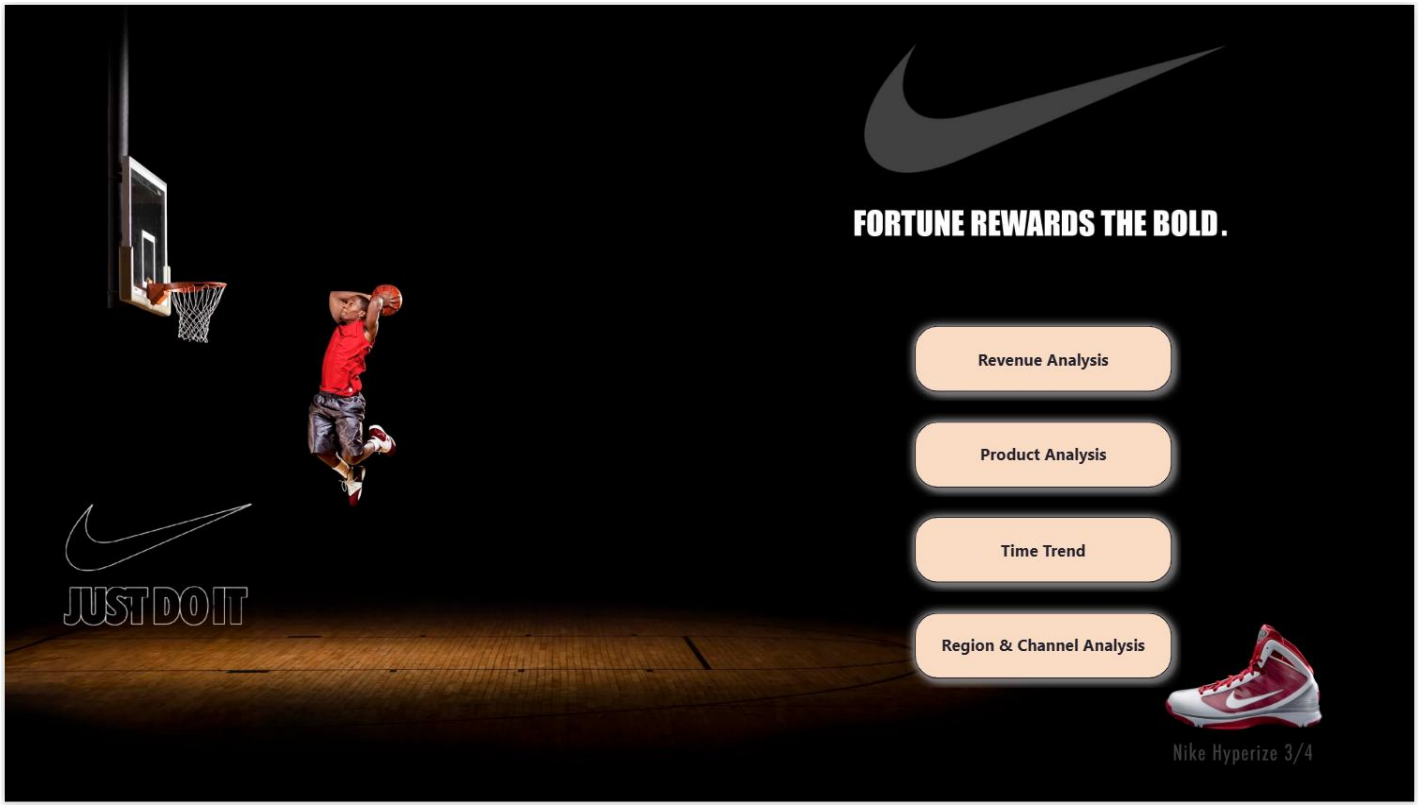
HOW WAS THE DATASET COLLECTED

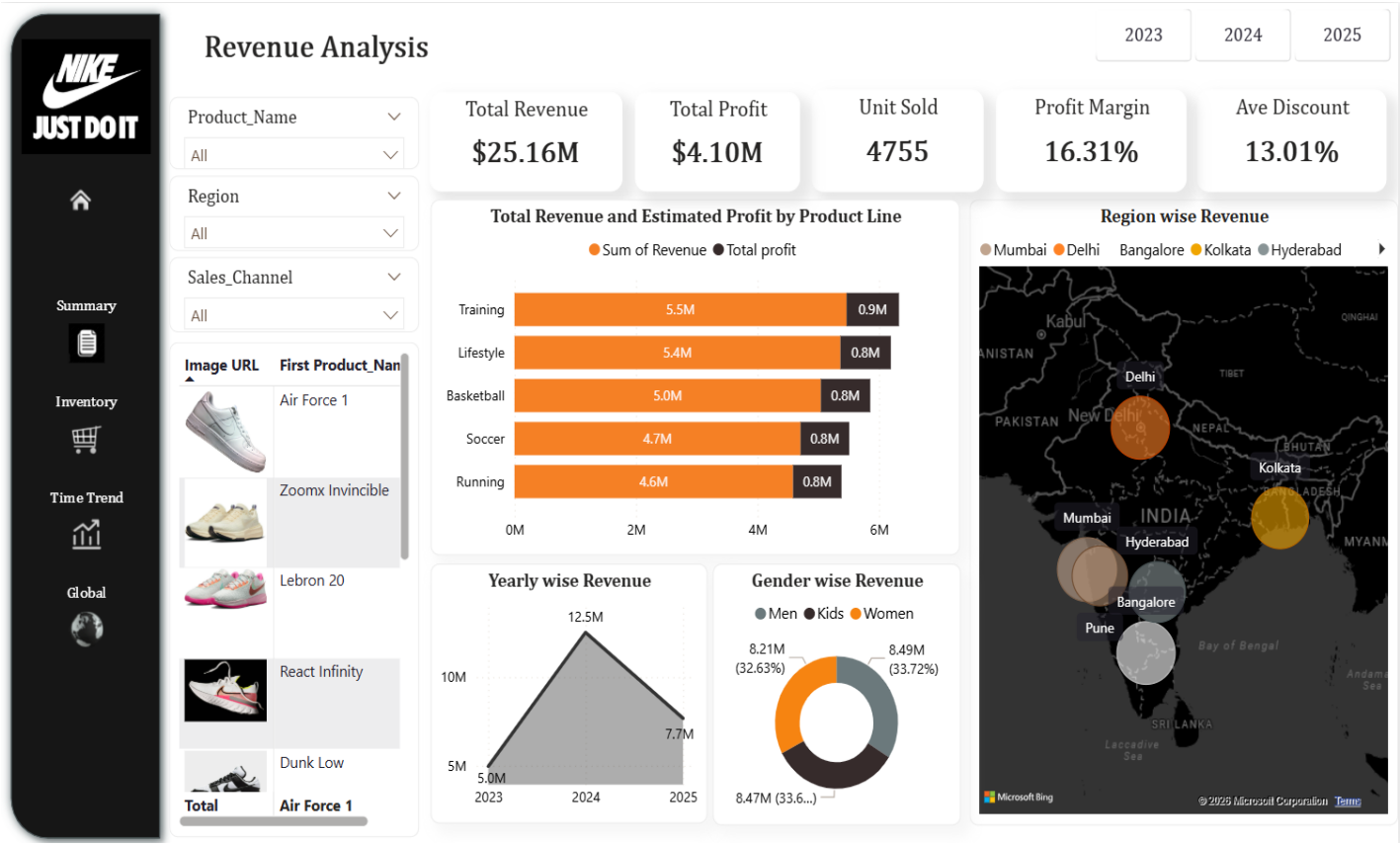
The dataset was collected from Kaggle Datasets. This dataset contains Nike retail and online sales transactions generated to simulate real-world business scenarios. It includes detailed information about product purchases, customer segments, pricing, discounts, revenue, and profit across different regions and sales channels. The dataset is synthetic but designed to closely resemble actual retail sales data, making it suitable for data cleaning, analysis, and dashboard development.

FEATURES IDENTIFIED FOR ANALYSIS

The dataset used for this project contains comprehensive information about Nike's sales performance, including product lines, gender categories, sales channels, regional markets, pricing, discounts, units sold, revenue, and profit. These features help analyze customer purchasing behavior, product performance, and overall business profitability. The primary objective is to evaluate sales trends, identify high-performing segments, and determine the factors that influence revenue and profit.

The first step in the pre-processing stage involved loading the dataset. Necessary Python libraries such as NumPy, Pandas, Matplotlib, and Seaborn were imported for data manipulation and visualization. After downloading the dataset as a CSV file, it was loaded into a Pandas DataFrame to initiate data cleaning and exploratory analysis. Before proceeding further, each column was carefully examined to understand its meaning, data type, and relevance to the analysis. Summary statistics, unique values, and sample records were reviewed to gain a clear understanding of the dataset's structure and characteristics.


FORTUNE REWARDS THE BOLD.
Revenue Analysis
Product Analysis
Time Trend
Region & Channel Analysis
JUST DO IT
Nike Hyperize 3/4


NIKE
JUST DO IT
Summary
Inventory
Time Trend
Global
Revenue Analysis
2023
2024
2025
Product_Name
All
Region
All
Sales_Channel
All
Image URL
First Product_Nan
Air Force 1
Zoomx Invincible
Lebron 20
React Infinity
Dunk Low
Total
Air Force 1
Total Revenue
$25.16M
Total Profit
$4.10M
Unit Sold
4755
Profit Margin
16.31%
Ave Discount
13.01%
Total Revenue and Estimated Profit by Product Line
Sum of Revenue
Total profit
Training 5.5M 0.9M
Lifestyle 5.4M 0.8M
Basketball 5.0M 0.8M
Soccer 4.7M 0.8M
Running 4.6M 0.8M
0M
2M
4M
6M
Region wise Revenue
Mumbai
Delhi
Bangalore
Kolkata
Hyderabad
Kabul
Delhi
New Delhi
Kolkata
Mumbai
INDIA
Hyderabad
Bangalore
Pune
Yearly wise Revenue
12.5M
10M
5M
5.0M
7.7M
2023
2024
2025
Gender wise Revenue
Men
Kids
Women
8.21M (32.63%)
8.49M (33.72%)
8.47M (33.6...)

Product Analysis

2023 | 2024 | 2025

Region: All

Basketball

Total Revenue	Total MRP
$25.16M	$15.19M

Ave Discount Rate	Total Profit
13.01%	$4.10M

Air Force 1

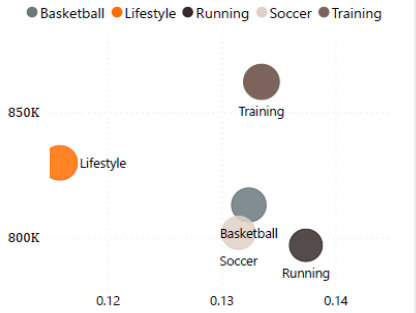
Product Name

Sale Channel Info

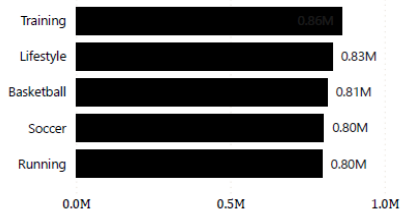
Online
75,85,223.06 Sum of MRP
1,25,82,601.91 Sum of Revenue
20,54,576.07 Total profit

Retail
76,02,424.35 Sum of MRP
1,25,82,076.75 Sum of Revenue
20,49,707.83

Relationship of Ave Discount Vs Profit



Revenue by Product_Line



Time Trend and Analysis

Online | Retail

Product_Name: All

Region: All

Order_Date: 26-07-2023 – 22-07-2025

Total Revenue	Total Profit	Profit Margin	Qty of Sold
$25.16M	$4.10M	16.31%	4755

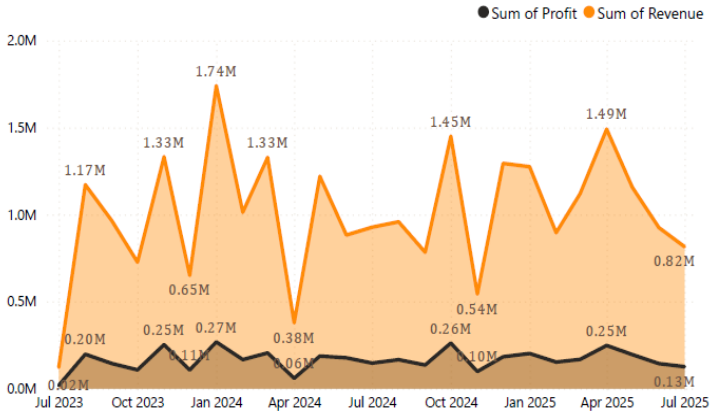
Air Force 1
12,49,431.23 Sum of Revenue
1,98,808.68 Total profit

Air Jordan
14,00,368.00 Sum of Revenue
2,00,150.78 Total profit

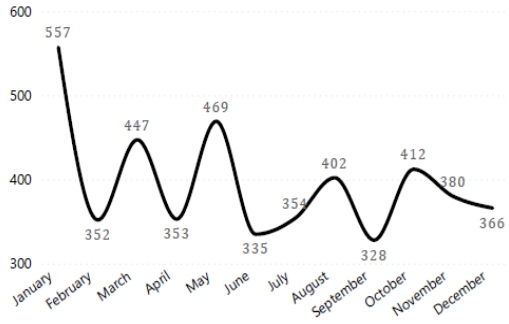
Air Zoom
10,41,258.54 Sum of Revenue
1,67,182.23 Total profit

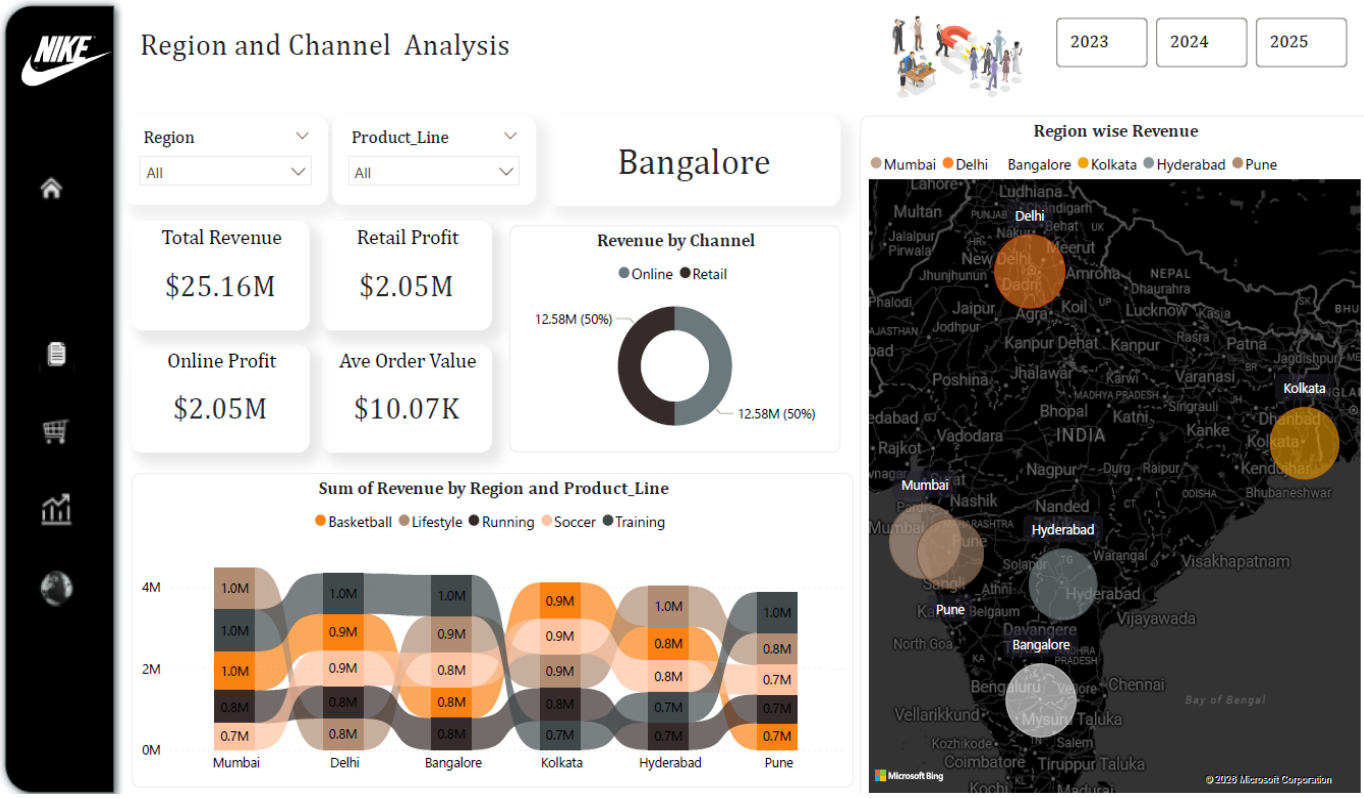
Blazer Mid
13,57,178.56 Sum of Revenue
2,21,051.18 Total profit

Monthly Wise Revenue and Profit



Monthly wise Qty of Sold





SUMMARY

The overall sales analysis shows strong and consistent growth in Nike's retail and online business, generating a total revenue of **$25.16M** with **$4.10M profit** and a **16.31% profit margin**. Sales performance varies across regions, product lines, and channels, with certain categories and metro cities contributing significantly higher revenue. Online and retail channels contribute almost equally, ensuring balanced growth. While moderate discounts increase sales volume, excessive discounts negatively impact profitability. These insights help identify high-performing segments and areas that require strategic improvement.

KEY INSIGHTS

1. **Regional performance varies significantly.**
   Metro cities such as Mumbai, Delhi, and Bangalore contribute the highest share of revenue compared to other regions.

2. **Discount levels strongly affect profit.**
   The average discount of 13% supports sales growth, but higher discounts reduce profit margins.

3. **Product lines drive different profitability.**
   Training and Lifestyle categories generate the highest revenue, while premium products deliver better margins.

4. **Online and Retail channels perform equally.**
   Both channels contribute nearly 50% of revenue each, showing balanced sales distribution.
5. **Time trend shows steady growth.**
   Revenue increased year-over-year, with 2025 showing the highest sales, indicating business expansion.
6. **Few products generate most revenue (Pareto effect).**
   Top-selling products like Air Jordan and Air Force 1 contribute a major portion of total revenue and profit.

FINAL SUMMARY

This dashboard provides a comprehensive view of Nike's sales performance across products, regions, and channels. It highlights revenue distribution, customer purchasing trends, and profitability patterns to identify the company's core growth drivers. The analysis shows that certain regions, product categories, and online channels contribute significantly more to total sales.

By examining pricing, discounts, and seasonal trends, the dashboard helps understand how business decisions affect profit margins. Overall, the insights enable better planning, smarter promotions, and more effective resource allocation to improve long-term performance.

CONCLUSION

In conclusion, the analysis demonstrates that strategic product focus, controlled discounting, and strong online presence are the key drivers of Nike's sales growth and profitability. High-performing regions and categories generate most of the revenue, while inefficient discounts reduce margins. By leveraging data-driven decisions and targeted marketing, the company can enhance customer satisfaction, increase profits, and achieve sustainable business growth.